

Tiange Liu

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EDUCATION

Ph.D. in Statistics, University of Washington <i>GPA: 3.82/4.00</i>	Sept 2025 - Ongoing
M.Sc. in Statistics, University of British Columbia <i>Supervisor: Dr. Trevor Campbell; GPA: 92/100</i>	Sept 2023 - Aug 2025
B.Sc., University of British Columbia <i>Combined Honours in Mathematics and Statistics; GPA: 90/100</i>	Aug 2019 - April 2023

RESEARCH EXPERIENCE

Research Assistant supervised by Dr. Xinwei Shen <i>UW Statistics</i> ◦ Developed experimental pipelines for network-based dimension reduction algorithm across synthetic and real-data settings ◦ Reproduced and extended benchmark experiments ◦ Conducted baseline comparisons with state-of-the-art nonlinear SDR methods	Feb 2026 - Ongoing
Research Assistant supervised by Dr. Trevor Campbell <i>UBC Statistics</i> ◦ Developed autonomous MCMC kernels, AutoStep, that automatically select appropriate hyperparameters at each iteration, and provably addresses the irreducibility issue via a stochastic step size selector and a Metropolis-Hastings correction ◦ Designed and implemented AutoStep sampler ◦ Conducted real data simulations on high-performance computing platform	May 2024 - Aug 2025
Research Assistant supervised by Dr. Jonathan Huggins <i>BU Statistics and Data Science</i> ◦ Applied AutoStep and parallel tempering algorithm to mutational biosignature problems	June 2025 - Aug 2025
Research Assistant supervised by Dr. Natalia Nolde <i>UBC Statistics</i> ◦ Collaborated with Scotiabank Risk Management Department, researched on creating a flood map in the drainage area of Squamish River and assessing the economic loss due to flooding events with various return periods under several climate change scenarios ◦ Used L-pareto process to fit and simulate the spatio-temporal distribution of precipitation data for the past 40 years ◦ Wrote a literature review on Intensity Duration Curve; wrote a project report on data analysis and assumption validation of the precipitation data	May 2022 - Aug 2022

PUBLICATIONS

Liu, T., Surjanovic, N., Biron-Lattes, M., Bouchard-Côté, A., & Campbell, T. (2025). *AutoStep: Locally adaptive involutive MCMC*. In *Proceedings of the 42nd International Conference on Machine Learning*.

TEACHING EXPERIENCE

Teaching Assistant

Sept 2025 - Ongoing

UW Statistics

- STAT 220: Statistical Reasoning (Summer, 2026)
- STAT 221: Statistical Concepts and Methods for the Social Sciences (Fall 2025, Winter 2026, Spring 2026)

Teaching Assistant

Sept 2021 - May 2025

UBC Statistics, Mathematics, and Computer Science

- STAT 302: Introduction to Probability (Winter 2024, Spring 2025)
- STAT 305: Introduction to Statistical Inference (Summer 2024)
- STAT 251: Elementary Statistics (Spring 2024)
- STAT 300: Intermediate Statistics for Applications (Winter 2023)
- STAT 200: Elementary Statistics for Applications (Winter 2022)
- MATH 104: Differential Calculus with Applications to Commerce and Social Sciences (Spring 2022)
- CPSC 107: Systematic Program Design (Spring 2022)
- MATH 103: Integral Calculus with Applications to Life Sciences (Winter 2021)

HONORS AND AWARDS

Graduate Trainee of Advanced Machine Learning Training Network (2024-2025): funded to UBC graduate students whose research focuses on machine learning

Faculty of Science Graduate Award (2023): offered to a full time graduate student on the recommendation of Faculties or graduate programs

Margaret Wylie Memorial Scholarship in Statistics (2023): offered to a woman who demonstrates excellence in the study of statistics and is beginning a graduate program in Statistics

Stanley W Nash Medal in Statistics (2023): offered to the most outstanding graduating student in the B.Sc. program in Statistics at UBC

Faculty of Science International Student Scholarship (2020-2022): offered to continuing international students in the Faculty of Science, UBC Vancouver, on the recommendation of the Faculty of Science

Trek Excellence Scholarship for Continuing Student (2020-2021): offered to the top 5% undergraduate students at UBC Vancouver

J Fred Muir Memorial Scholarship in Science (2020-2021): rewarded to students in the Faculty of Science on the recommendation of the Faculty

Science Scholar (2020-2021): awarded to students achieving greater than 90% average at UBC

Outstanding International Student Award(2019): offered to outstanding international students entering undergraduate programs at the University of British Columbia's Vancouver campus

PROFESSIONAL EXPERIENCE

Statistical Consulting

Jan 2024 - May 2025

UBC Statistics

- Evaluating the Impact of the Adolescent Health Program on Sexual and Reproductive Health Literacy Among Adolescent Girls in Rural Nepal
- Meta-Analysis of Endovascular Techniques for False Lumen Thrombosis in Type B Aortic Dissections
- Statistical Assessment of Risk Factors for Bleomycin-Induced Hyperpigmentation in Low-Flow Vascular Malformation Patients
- Statistical Advice On Investigating the Outcome Trajectories Following Cerebral Venous Thrombosis

- Understanding Dietary and Physical Activity Traits in Breast Cancer Survivors
- What is the best way to promote zero calories beverages?

SKILLS SUMMARY

Programming languages: Julia, R, Python, C++, MATLAB, Java, Dr. Racket

Database: Oracle SQL

Tools: GitHub, L^AT_EX, HPC, SSH, Nextflow, Jupyter, Markdown, Microsoft Office

Certificates: Coursera official certificates for Deep Learning and Machine Learning at Stanford University